

Dr. Santosh Todkar

AI/ML Engineer | Bangalore | +91-7744004068 | todkarsant@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Summary -

AI/ML Engineer with 3.5+ years of experience developing, deploying and monitoring Machine Learning and Generative AI solutions across BFSI, Telecom and Industrial domains.

Delivered ML and AI solutions that reduced customer churn by 33%, improved inventory replenishment efficiency by 25%, and reduced document validation time from 2.5 hours to 15 minutes.

Experience - (3 years 6 months)

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| 1. AI Engineer, Acronotics Limited | September 2025 to date |
| 2. Data Engineer, Reach Mobile (Datami Mobile Solutions Pvt Ltd) | February 2023 to July 2025 |
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Core Skills -

Programming & Data: Python, SQL, PySpark

Machine Learning & NLP: XGBoost, PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, NER

Generative AI: LangChain, LangGraph, RAG, AI Agents, Prompt Engineering, Tool Calling, MCP

Cloud & Platforms: AWS (S3, DynamoDB), Azure AI Foundry, Azure App Services, Azure ML workflows, Oracle Cloud Infrastructure, GCP (BigQuery), Databricks

Software Engineering: Git, FastAPI, REST APIs, Docker, CI/CD, MLFlow, Event-driven Architecture

ML Expertise - NLP, Feature Engineering, Predictive Modelling, Model Evaluation & Explainability (AUC-ROC, Precision/Recall, F1, SHAP, LIME), Hypothesis Testing, Model Monitoring & Drift Detection, MLflow, AutoML

GenAI Expertise -

Advanced RAG Technologies - Azure AI Document Intelligence, Hybrid Retrieval, Semantic Search, Cross-Encoder Reranking (bge-reranker), Prompt & Retrieval Caching

LLM Evaluation & Responsible AI - Azure evaluators, Groundedness, Similarity, Relevance scoring, prompt validation and trusted AI practices, RAGAS

LLM Observability - LangSmith, Human-in-the-Loop, Hallucination Detection, Context Engineering,

Machine Learning & NLP Solutions -

Reach Mobile - (Telecom analytics company supporting US mobile operators through predictive analytics.)

A. ML-NLP Powered Customer Retention Platform

1. Developed an NLP-powered customer retention model that reduced churn by 33%, enabling more targeted retention campaigns and improving customer lifetime value.
 2. ML pipeline deployment - Automated NLP feature engineering using topic modelling, sentiment analysis, and keyword extraction with Databricks real time pipeline.
 3. Model selection/ML Model ranking - eXtreme Gradient Boosting (best performance after comparison with Random Forest, Logistic Regression)
 4. Drift detection: Monitored model performance monthly, retrained if ROC-AUC dropped below 0.80.
 5. SHAP and LIME validation techniques were used for the model evaluation and explainability.
 6. Delivered predictive analytics solutions supporting customer retention and operational decision-making.
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B. ML Inventory Forecasting Platform

1. Built an XGBoost-powered forecasting platform that improved inventory replenishment efficiency by 25% through automated stock prediction and threshold-based alerts.

2. Conducted exploratory data analysis, feature engineering, and data validation to improve predictive model performance and support production deployment.
 3. ML pipeline deployment - Data collection from AWS S3 bucket and DynamoDB. Preprocessing, training, validation, and deployment hosted on Databricks.
 4. Showed predicted V/s Actual SIM stock over time - eXtreme Gradient Boosting regression
 5. Inventory alert system – Highlights low stock or high stock based on thresholds and suggested EOQ for the replenishment
 6. Drift detection: Tracked model refresh cycles, last retrain date, and performance drift
 7. Collaborated with cross-functional teams and mentored interns to translate analytics solutions into measurable business outcomes.
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Enterprise GenAI & AI Automation Solutions –

Acronotics Limited – (Enterprise AI startup delivering GenAI solutions for banking and industrial clients.)

A. Banking Document Intelligence AI Validation Platform

1. Built and deployed a RAG-based document extraction and validation platform using Azure OpenAI for automated Letter of Credit document processing.
 2. Reduced Letter of Credit document extraction & validation time from 2.5 hours to 15 minutes.
 3. Designed asynchronous AI workflows using Celery and Redis, enabling low-latency document processing and scalable concurrent execution to process 30 documents per day as batched pipeline.
 4. Led end-to-end delivery of AI solutions, collaborating with architects, business stakeholders, and mentoring junior engineers from solution design through production deployment.
 5. Designed prompt and retrieval strategies that improved reasoning quality, groundedness, and extraction accuracy for enterprise financial documents.
 6. Developed production-grade FastAPI microservices enabling enterprise AI integration.
 7. Implemented hybrid retrieval combining vector search, semantic ranking, and LLM evaluation to improve extraction accuracy and reduce hallucinations.
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B. AI Contract Price Intelligence Application

1. Architected a multimodal RAG system using Azure OpenAI and Azure AI Document Intelligence for intelligent contract price benchmarking across unstructured documents.
 2. Implemented hybrid retrieval using vector embeddings, semantic search, and BGE Cross-Encoder (bge-reranker) to improve retrieval precision and contextual relevance.
 3. Built stateful LangGraph-based agentic workflows with tool-calling capabilities and transformer-based document intelligence for multi-step reasoning, contract analysis, and GenAI-assisted price benchmarking.
 4. Optimized production inference through prompt and retrieval caching, reducing latency and LLM token consumption for enterprise workloads.
 5. Collaborated with business stakeholders, architects, and junior engineers to deliver and continuously optimize AI workflows.
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Research & Publications –

1. Doctoral research background in composite materials and sustainable biocomposites.
 2. Research direction: ML-assisted material characterization, surrogate modelling and multi-objective optimization for sustainable engineering materials.
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AI Certifications –

1. Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate.
2. AMD AI Developer Academy - Reinforcement Learning with Large Language Models.
3. Introduction to Claude code, Claude cowork, Claude 101